SG-258-250

Precision Industries Testing Department	GENERATING-SET LOAD TEST	Model No: PI 250 C	60 Hz
		Type: Close	Job No: 9053
		Color: Orange	Date: 07/03/2017

Serial Number:								
PI Model No:	PI 250 C	Prime Power:	225	[kVA]	P.F (Load):	1	Voltage:	380

Generating Set Details:

Engine:	Cummins	Model No: QSB7 - G5	Serial No: 84276149
Alternator:	Stamford	Model No: UCI274H1	Serial No: N17F240046
Controller:	COMAP	Model No: AMF 9	Serial No: 16024ADC
C. Breaker:	Metasol 630A	Model No: ABN 803C	Serial No: N/A
C.Transformer:	Lovato 800/5A	Model No: N/A	Serial No: N/A
AVR:	Stamford	Model No: SX 460	Serial No: 111760 - 3552
Cooling System: Water Cool		Enclosure Type: Canopy	Protection Class: IP23

Load Test Ambient Conditions:

Temperature	24	[°C]	Altitude:	0	[m]	Humidity:	78 %

Generating Set Rating:

Testing Full Load Current:	273.48	Amps	Nominal Current:	341.9	Amps
Power Correction Factor (De-rating):	0.000	%	Corrected Prime Power:	0.0	[kVA]

LOAD TEST

Load	Run Time	Freq	Speed	VOLTAGE L-L			CURRENT			Power	Coolant Temp	Oil Pressure	Charg-Alt.
				L1	L2	L3	L1	L2	L3				
%	min	Hz	RPM	V	V	V	A	A	A	kW	C	bar	VDC
0 %	5	60	1800	380	380	380	-	-	-	-	52	4.80	14.1
25 %	5	60	1800	380	381	381	68	69	69	45.0	76	4.40	14.1
50 %	5	60	1800	380	381	381	134	134	133	87.0	80	4.12	14.1
75%	5	60	1800	380	381	382	204	204	203	133.0	84	3.92	14.1
100 %	10	60	1800	380	381	382	273	273	272	180.0	86	3.68	14.1
110 %	10	60	1800	381	382	382	302	302	301	199.0	86	3.60	14.1

LOAD ACCEPTANCE TEST

Suddenly Block Load %	Voltage [V]	Frequency [Hz]	Currnet [A]	Response Time [sec]
50%	380	60.0	134	0
0%	384 - 380	61.4 - 60.0	0	2
50%	376 - 380	58.8 - 60.0	136	2

PRECISION INDUSTRIES
TESTED
Testing Department

PROTECTION SYSTEM TEST

Over Frequency SD	Under Frequency SD	Over Voltage SD	Under Voltage SD	Oil Pressure SD	Coolant Level SD	Coolant Temp SD
115%	85%	460	340	1.0 ECM	ECM OK	110 c

ACCESSRIES DETAILS

	Model:		Serial No.	
	Model:		Serial No.	

PI Document no.	PI-TST-LT-01	Rev.	01

PO BOX NO. 17, BARNACK ROAD, STAMFORD LINCOLNSHIRE, PE9, 2NB, ENGLAND.

STAMFORD

Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N17F240046**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: UCI274H1	**AVR:** SX460	**AVR NO:** 1117603552	**ENCLOSURE:** IP23	
RATED KVA: 200	**RATED VOLTS:** 400	**RATED AMPS:** 288.7	**FREQUENCY:** 50	**PHASE:** 3
POWER FACTOR: .8	**STR CONN:** Series Star	**STR WINDING:** 311	**DUTY:** Base Continuous	**WIRE:** 4-Wire
EXC. VOLTS: 58	**EXC.AMPS:** 2.3	**AMB TEMP:** 40		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.03371	2+	OK	H
ROTOR	1.9663	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	12	.5	0.00
50	400	288.7	200	.8	160	58	2.3	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.UC27.4.H.1.311.,2,.,11.5,.SX460.Black - RAL9011
Remarks:

Date: 14-JUN-17	Job No: 3484504-007	AUTHORISED BY **Manoj K Upadhyaya** Quality Assurance

Certificate Of Conformity

Quality System Approval
ISO 9001/2008

Frame Size : **UCI274H1**

Serial Number : **N17F240046** **Manufactured Date :** 14-JUN-17

KVA: **200**

Authorized By Quality Assurance: **Manoj K Upadhyaya**

This document is electronically generated **Print Date: 14-JUN-17**

EU DECLARATION OF CONFORMITY

Cummins Generator Technologies

This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

2014/35/EU	Low Voltage Directive
2014/30/EU	The Electromagnetic Compatibility (EMC) Directive

and that the standards and/or technical specifications referenced below have been applied:

EN 61000-6-2:2005	Electromagnetic compatibility (EMC). Generic standards - Part 6-2: Immunity for industrial environments
EN 61000-6-4:2007+A1:2011	Electromagnetic compatibility (EMC). Generic standards - Part 6-4: Emission standard for industrial environments
EN ISO 12100:2010	Safety of machinery - General principles for design - Risk assessment and risk reduction
EN 60034-1:2010	Rotating electrical machines - Part 1: Rating and performance
BS ISO 8528-3:2005	Reciprocating internal combustion engine driven alternating current generating sets - Part 3: Alternating current generators for generating sets
BS 5000-3:2006	Rotating electrical machines of particular types or for particular applications - Part 3: Generators to be driven by reciprocating internal combustion engines - Requirements for resistance to vibration

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Limited, 49/51 Gresham Road, Staines, Middlesex, TW18 2BD, U.K.

Signed: Date: 01st February 2016	Name, Title and Address: Kevan J Simon Global Technical and Quality Director Cummins Generator Technologies Fountain Court Lynch Wood Peterborough, UK PE2 6FZ

Description UCI274H1

Serial Number N17F240046

450-16383-F